



Canadian Ocean Modelling Forum workshop: 11-13 May 2026

Attendees: Day 1- 54 online, Day 2- 36 online, Day 3- 46 online

Please see workshop slide for session introduction information

Workshop slides: [PDF NEMO workshop 2026-Slides.pdf](#)

1. Ocean model development: Globally and in Canada

(Discussion leads: Paul Myers and Nicolas Lambert)

- Monday 11 May, 10:00 am MDT

- Paul provides the background of the CoP
- Spatial and temporal scale discussion:
 - **Spatial**
 - Greg Smith- Good to think beyond just NEMO. One helpful thing would be to think about how we can facilitate the sharing of the products. We can go download it, but there is knowledge about how to make it useful. Would it be useful to have Slack or a tool to help share knowledge?
 - Paul- make files available and what impacts or work with making things available?
 - Amber- We need better data products, especially rivers. A potential **GitHub organisation** to help provide links, tools or methods repository.
 - Paul- not a link to data but the methods to get the data on the correct points.
 - Andrew Peterson- we should link the intercomparison groups that have developed tools. For global and regional models.
 - Paul- take advantage of the knowledge of expertise of different spatial scale expertise to help assess your larger scale models.
 - Nicolas- It would be helpful to have information on the different global-to-regional model scales, and to have clear communication for crossing those boundaries to different scales.
 - Paul- You could pull data from the sites, e.g., ERA, but there are some errors that occur if you use it off the shelf, and it would be good to communicate who has done what. NEMO has different atmospheric bulk forcing schemes, and most use the standard ones, but is there a need to explore the different bulk forcings?
 - Greg- Have done some different scheme investigations, but depends on what you want to use. COARE is the most up-to-date system, but it doesn't mean it will be the best results for your use
 - Neil Swart- Jonathan has done some work on the different bulk atmospheric forcing. If you go down from 1 degree to 1/12th degree and you do all the changes to get it to downscale, we don't know where all the differences are coming from. Testing was done on a global $\frac{1}{4}$ and to a regional $\frac{1}{4}$ to see which is from the regional configuration, and found atm

- applications have an impact.
- Paul- It would be good to have the knowledge shared, as there are tests that have been done but not necessarily published, and it is useful for everyone to see what changes across the different tests. The sensitivity testing sharing?
 - Neil- Communication across scales: the history of CCCma takes a single model and applies it at different scales, even over time; use as a common basis, as there is a well-tested system to apply. There has been a lot of diversity in a very small community, and would it be beneficial to think about how to reduce the diversity and increase efficiency?
 - Amber- The code base makes it more challenging and where the effort goes. How do we reconcile having one code base for the larger to smaller scale applications?
 - Greg- Code is getting easier, maybe not BGC, but the physical is getting better through code sharing and branching (Configuration and Canadian 3 oceans system). Namelist sharing would be useful for parameter sharing and what has been done and tested and the results of that shared.
 - Stephanie Taylor - A consistent codebase does become tricky with different versions. Namelist could be an achievable low-hanging fruit.
 - Nicolas- Having the same codebase might be the easiest, using a Git merge and leveraging it more efficiently. Maybe those protocols need to be added to ensure that if a code parameterisation is added, you can switch it on and off.
 - Stephanie- Certainly version control helps to streamline it, but if you have one system that is SI3 and others that are using LIM2 and LIM3, how do we start merging the different usages. How do we manage that migration to a consolidated system?
 - Greg- Should we have a discussion on how to have a version fork and version git in the consortium? But we can still see the differences from the different groups.
 - Neil Swart (chat) "In the UK they have organisations like NERC that seem to facilitate collaboration on "common configurations" and tools etc". Similar mechanism in Canada?
 - Andrew- If we want to be part of the NEMO consortium, this will need to be achieved.
 - Amber- Unifying domains, different institutes are using different versions. Eg. Government is v4 and 4.2 and then academia in version 5. How do we have that collaboration?
 - Paul- Learning a lot from CanTODS, work to get the domains close, but it is the forcing and the namelists that are being shared. eORCA twin experiments at higher resolutions will be run to see change in resolution on domain.
 - Stephanie- Do we need to be an official consortium contributor to be able to contribute to the consortium?
 - Paul- Stability is not the biggest question; had many issues with version 4.2 over version 5 and likes version 5 more.

- Michael - who has the capacity to take this on to get it done? Questions on engineer time to get it done? Understanding that it may be too costly to get it to consolidate, and maybe it would be useful to bring the code to a more similar code base.
- Paul- Should we be applying for more money to get the code consolidated?
- Geoff- We should start with having our code bases public-facing and share-friendly. People can talk more easily to one another. Post your namelist; they evolve all the time. The community helps to facilitate the discussion and sharing of information on why changes have been made.
- Fred Dupont- You can join the consortium working groups.
- Paul- Forcing products to be able to run for most recent years. If you collaborate with observationalists? It is harder to get the most recent periods. Hindcasting (1950s to present). What do people do for the last three years for icebergs or rivers to avoid issues?
- Stephanie- Some automatic scripts that grab gauge data. The DBNM group collects the operations output for updating. It is for 3 gauges on a regional scale. Can it be scaled up?
- Paul- International domains are the issue.
- Greg- The real issue is with data from the North. Would be good to reach out to hydrology groups and see where they are at for the Arctic.
- Michael- HRDLPS for west coast. 2019 to present-ish.
- Natasha- For future CMIP data can be used, but would it be possible to help with the hindcast? It is a prediction forcing but might be more useful for some years?
- Duo Yang- It is not necessary to be perfect.
- Nicolas- **Diagnostics discussion**
 - There is a WG at the NEMO Consortium that is focused and interested in diagnostic tools and sharing.
 - Stephanie- Keen to share the code, but a lot of the code has been developed for specific needs. NEMOCookbook is more versatile, and the code developed by our group is for more nearshore work. Has participated in the WG and intends to participate more.
 - Nicolas will share the link to the WG
 - Greg- If there were a central way to share things. E.g., a GitHub page with code, but then also an analysis section and with links to publications? Would that be useful to the group?
 - Paul- CDFTools is a very good toolbox. Coordination of tools and code even within the groups is very difficult, and bringing it onto a larger CoP scale may be a very large effort. Simple PDF guides that could be made available, e.g., Xianmin tripolar grid velocity shift tool, might be a good first step forward that could help.
 - Nicolas- Good to share the tool, but there is a difficulty in terms of usage side. The architecture of the code, the code base... but we want to still use and share. Code needs to come with a README.
 - Krysten Rutherford - "From Elise Olson

<https://github.com/SalishSeaCast/tools>”

- Stephanie- For coordinating, we do risk splitting discussions. There are internal gits, but then there are public-facing ones. How do we make it all public-facing while ensuring standards?
- Greg- If we do have different git repositories, how do we manage the different histories? There hasn't yet been a discussion on how to put AI into use, and if we make it available to AI agents. Can we use that to make code more accessible and transparent and transferable?
- Nicolas- public-internal strategy, have internal and then a public push with a main approved job pushed onto the public facing git.
- Amber- GitHub offers a GitHub Organisation that can be used to help have a centralised point, as it is free for academic institutions. Do we need the internal repo to keep it separate from the public facing?
- Nicolas- It is more for the security system and concerns. And a discussion around what is on the main approved push that will then will be made public
- Amber- What needs to be public?
- Nicolas- The decision-making takes a lot of effort.
- Paul- Useful things are Lagrangian particle models; can we coordinate that? If you are evaluating the models, you need to compare to observational data, and we need to communicate with observationalists that know the data setup to help. Link more with CIOOS and DFO collaborations.
- Stephanie- Coordination of data between different groups and has done work with standard formats of data and needs to be coordinated. Each group had different priorities, and that is usually focused on when collaborating.
- Greg- There was an Ocean Predict meeting and Copernicus meeting. Need standard diagnostics and guidance or SOPs on how emulators are trained. GitHub could be used if we can share benchmarks to make sure you are doing things correctly.
- Hatem- Opendrift is used by the Canadian Coast Guard and is covering all ports and would be keen to collaborate on incidents and Lagrangian particle tracking studies.
- Nicolas- **Adjacent modules and libraries**
 - Jim- It is important to know what the CCCma has done in the past to collaborate and coordinate the BGC modelling. Built around the PISCES framework that was then hollowed out and then run. Has caused issues with new versions where there will be PISCES, but it won't be used. Learn to keep the PISCES code base and don't tamper with it, and build and couple CMOC and CanOE in the new versions but with its own architecture. We need to learn from the past, and no need to redo it every time with code updates.
 - Nicolas- CICE issues but still using NEMO instead of LIM or SI3? Will CICE still be used in the future?
 - Fred- (For CICE) Still needed to share MPI I/O's, and then tides were an

issue, and we can't use AGRIF as it is outside the NEMO framework and support with CICE.

- Nicolas- Is the future strategy to still use CICE or move?
- Fred- We don't have the resources to focus on 2 codes, so there will be a code switch to SI3. CICE is a great model but no resources.
- Nicolas- The coupler is an issue. Not using OASIS at the moment, but may need to switch. Having issues with resolution coupling. There has been ML development to use OASIS for the coupler and implementation in the NEMO code.
- Neil- Couplers are an area that we need to collaborate more on. Will OASIS become redundant when the main developer retires?
- Nicolas- Other examples of different modules and models? Eg. rivers (land-ocean), ice sheet dynamic model.
- Fred Dupont (chat) "I have good experience of coupling WW3 with NEMO via OASIS"
- SterlinJ (chat) "On using XIOS as a coupler: Towards New Community-Oriented Coupling Software In Climate Modeling: Evaluating XIOS Beyond I/O (https://hal.science/tel-05543302v1/file/Polimi_thesis.pdf)"
- Geoff- Some might be better to use offline than online with some experiments
- Greg- IRIS coupler development? Is there interest in getting involved?
- Nicolas- Definitely should collaborate with coupler and IO
- Neil- There is an opportunity to use the different couplers

Papers and resources:

- Canadian NEMO Forum Community of Practice: [Canadian institutions](#)
- DRAKKAR 2026 [public documents](#)

2. AI integration into NEMO modelling and analysis processes

(Discussion leads: Amber Holdsworth and Andrew Shao)

Intro slides from Andrew Shao-

https://drive.google.com/file/d/1kGGgTKs0KgVHG1mLdp_wHW5kUgwxx3_S/view?usp=drive_link

- Monday 11 May, 12:00 pm MDT

[Advised to read the second transcript of day 1 as many details are within the transcript that were missed in the meeting notes]

- Andrew Shao: **Broad overview**
 - Fred- We care about the first principles, but we don't have everything in the whole system down; we could use emulators to help resolve certain processes and problems. Also, wave action on the ocean, as a wave model, is very computationally expensive.
 - Andrew- We rely a lot on heuristic relationships, but AI models can be developed quickly and used to ask whether it is worth adding a particular

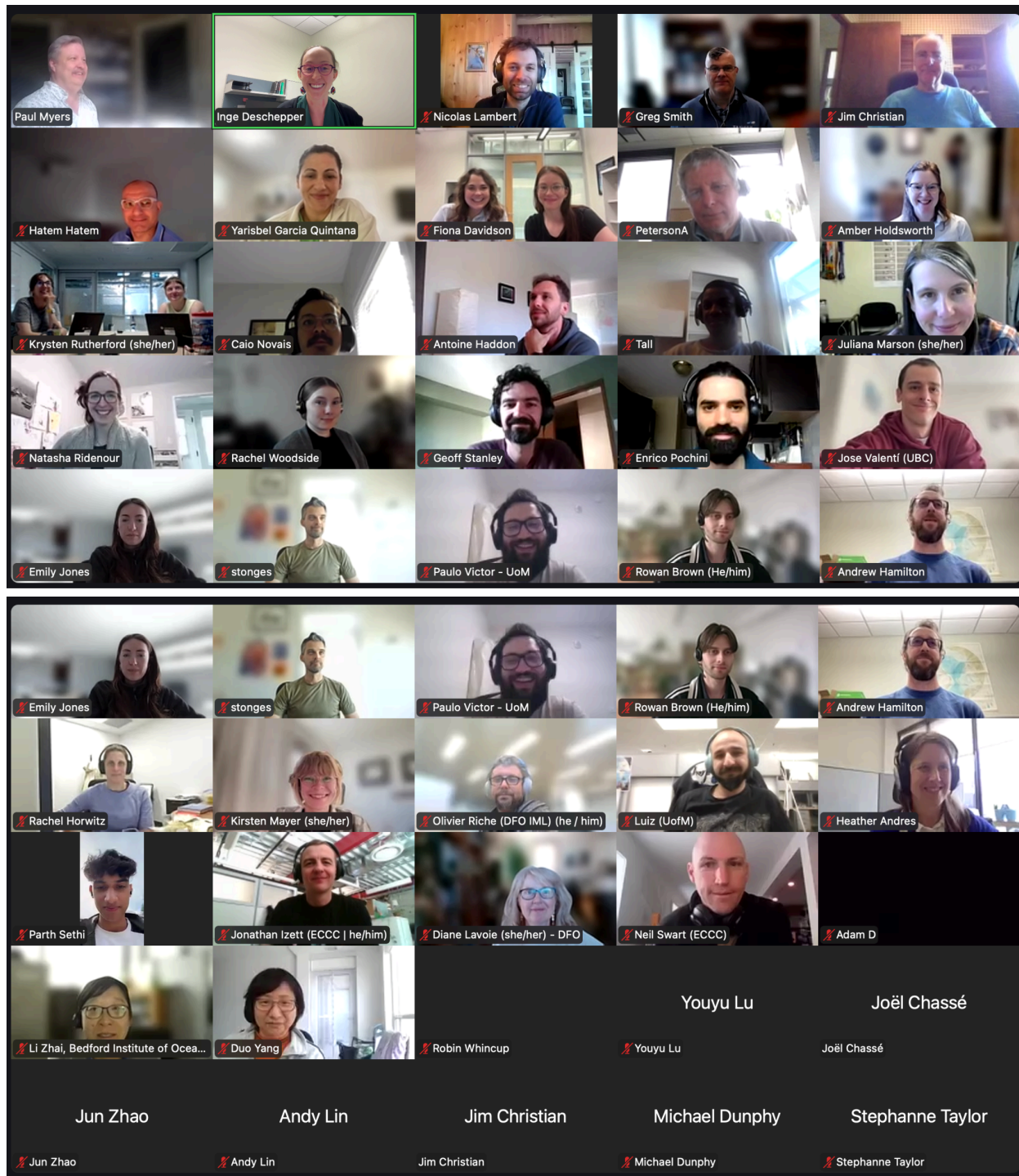
process in the initial hypothesis testing, and whether it is worth going down those paths.

- Amber- People have had this fear with AI replacing our models, and we will not have our jobs. Is there a part of the neural networks that could add a new beast into the computational modelling community?
- Andrew- Yes, there is probably a hybrid space in the middle that will bridge pure computational models and ML/neural networks. How to get there? And do it so that it is not only for computational modelling?
- Amber- Think of it as a tool for numerical modelling, but not a replacement.
- Andrew- Latent space. What is the reduced order to represent the system?
 - Amber- Adam Monahan's group using
 - Hatem- What is the dimensionality of the input database? What can you do to reduce the dimensionality? Hyperparameters optimisation?. Why LSTM and not ...?
 - Andrew- How do we make the problem simpler for machine learning problems to solve? How do we make sure we capture the next prediction snapshot but also long-term prediction? How do we make these models smaller so that the hyperparameterisation space is less sensitive?
 - Greg- Tides were the focus. For some things, the error growth becomes more important.
 - Hatem- dimensionality of database
 - Andrew- dynamic model decomposition, spectral proctoral orthogonal
 - Amber- what is an architecture? What is a model?
 - Andrew- Architecture is the equation of a line....(see transcript)
- AI/HPC research HPE labs (Andrew)
 - HPE labs perform fundamental research with external partners across computer and domain spaces
 - 3 main research areas
 - What numerical techniques can be applied to AI surrogates of dynamic systems?
 - Where are the benefits of blending AI methods and simulation to open new areas of research?
 - How do these hybrid simulation/AI apps challenge our view of computing, and what software needs to be developed?
 - Focus on open source science
 - How to connect Fortran with AI ...
- Paul- Lagrangian applications of AI/ML in the Hudson Bay
 - How to evaluate high-resolution models with sparse data observations? Is there value for AI/ML in evaluating high-resolution models?
 - Andrew- The question to the community is, what are the areas of challenge? Knowledge of techniques? Application of techniques? Tools?
 - Paul- All... Looking for patterns and exploration. What are some good approaches that can help analyse these very large datasets? Very open on how to do it. Integration and collaboration. Diagnostics and applications. As much as emulators can help with parameter testing.

- Hatem- Lagrangian tools used every day. OceanDrift from OpenDrift. Diffusivity coefficient from the trajectories. Using Monte-Carlo to analyze, but what are the right ensemble of parameters to get the right trajectories from a batch of 1000's. ML can be used to get the right combination of parameters for trajectories.
- Caio- How do you select the correct algorithm for the research?
- Hatem- There is no universal formula for selecting an algorithm. It depends on the type of database you have; find the point of highest variability and go from there. Only through experience will you learn which one to use when. Benchmark all the different models and see the variability on them. Be careful of overfitting. If you see 100% correlation, that is not right.
- Damien- I can write down challenges in my field of sea ice dynamics, which would be a great place for ML because sea ice rheology is non-linear, and needs a very expensive solver. The problem I faced with a colleague of mine a few years ago are: we could build a sea ice emulator, but we know our current models are wrong. Then, how to build a physics-consistent sea ice dynamical model that is not only a black box that reproduces sea ice movement but also informs us on the nature of sea ice mechanics... also we do not have that much data to train with, and almost no stress in situ data (which is what our momentum equation needs...)
- Andrew- Physics-informed neural networks, what is the correct architecture that is required to resolve them?
- At Drakkar, [Veros's main documentation](#) was presented as a Python-friendly version of oceanigans. Fully differentiable.
- Amber- Climate projections are built on very robust physics, but what about the biogeochemistry? We don't know if there is stationarity. What can this space be used for BGC modelling?
- Andrew- What is the most natural way of modelling the natural system? Rates and pools at the moment. The larger questions: probability distributions from first principles? Can we bring in a heuristic understanding of ML models to BGCM?
- Jim- it could be that ML could potentially be better for BGCM than the models, but how can you use the observations to constrain the model? The observations may be the big questions.
- Hatem- Find an empirical method that can explain your profile and then use that for the ML

Papers and resources:

- <https://www.sciencedirect.com/special-issue/10FCZL672Q7>
- Jang et al. 2026
 - <https://github.com/nicezheng/SSTODE-code>
 - <https://ojs.aaai.org/index.php/AAAI/article/download/41209/45170>



Day one participants from left to right:

Top: Paul Myers, Inge Deschepper, Nicolas Lambert, Greg Smith, Jim Christian, Hatem Hatem, Yarisbel Garcia Quintana, Tahya Weiss-Gibbins, Fiona Davidson, Adam Peterson, Amber Holdsworth, Krysten Rutherford, Elise Olson, Caio Novais, Antoine Haddon, Abdhul Tall, Juliana Marson, Natasha Ridenour, Rachel Woodside, Geoff Stanley, Enrico Pochini, Jose Valenti, Emily Jones, Stonges(?), Paulo Victor, Rowan Brown, Andrew Hamilton, Rachel Horwits, Kirsten Mayer, Olivier Richie, Luize Henrique

de Silva, Heather Andres, Parth Seth, Jonathan Izett, Diane Lavoie, Neil Swart, Adam D (?), Li Zhai, Duo Yang, Robin Whincup, Youyu Lu, Joel Chassé, Jun Zhao, Andy Lin, Michael Dunphy, Stephanie Taylor.

3. ECR day: AI training

(Kirsten Mayer [NCAR] and Andrew Shao [HPE], Dhamma k [NCAR]) (36 participants)
- Tuesday 12 May, 10:00 am MDT

Compute Ontarios summer school 2026-
<https://training.computeontario.ca/coss2026.php>

Recording is being made of the lecture and will be made available:

Google Slides of lecture done by Kirsten Mayer:
https://docs.google.com/presentation/d/1nDmbLUAAI9aLTBx4skg_yHToQGKGbD57SoK5fHHAtmY/edit?slide=id.p#slide=id.p

Exercise:

1. Open Collab: <https://colab.research.google.com/>
2. Click on the tab that says GitHub. This will allow you to paste the below GitHub link to get the file for the Collab
3. Paste the following link for GitHub for Collab:
<https://gist.github.com/ashao/e22a96557cb17df024f3a220865d1e2c>
4. Start playing

Other tutorial chapters from the NCAR GitHub:
https://github.com/NCAR/AIML-Tutorial/blob/main/Lecture1_ANNintro/ANN_intro.ipynb

Papers and resources:

- <https://arxiv.org/abs/2605.07896>



Day 2 participants from left to right at the end of the tutorial:

Andrew Shao, Inge Deschepper, Dhamma K, Amber Holdsworth, Kirsten Mayer, Stonges (?), Enrico Pochini, Fiona Davidson, Shuqi Lin, Jose Valenti, Tahya, Weiss-Gibbons, Thomas Guyndett, Grace Kirkpatrick, Caio Novais, Robin Whincup, Parth Sethi, Andy Lin, Tessa Sou, Debora Lucatelli, Diego Martinez, Robin Woodside

4. Biogeochemical models development and directions

(Discussion leads: Elise Olson and Inge Deschepper)

- Wednesday 13 May, 10:00 am MDT

BGC table - All, please fill in any updates or missing information since last year (29 May 2025)

Group	Model	What are you working on	What are you stuck on	Contact
University of BC	SMELT https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2019JC015766	Oxygen	Sediment impact on a sediment model	A. Tall
		Zooplankton		K. Suchy
		Carbon impact on plankton	Finding good studies that constrain at least two carbon system parameters	K. Suchy D. Ianson S. Allen
	Carbon			C. Stang

	(SKOG) https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021GB007024 Oxygen in draft form	Outfall/Alkalinity Enhancement Bacteria Turbidity River input		A. Tall J. Valenti E. Olson S. Allen
University of Alberta	BiGCIIM and BLING	Sympagic Carbon Upgrade from 3.6 to 5 Nutrient transports	Numerical instability due to tides in Foxe Basin Sediment burial fraction	Paul Myers Inge Deschepper
DFO (NL/Maritime)	CanOE	Mussel carrying capacity. Bivalve ecophysiology	For now, tuning the physical model	Thomas Gyuondet Andry Ratsimandresy
DFO/CCCMa	CanOE	tuning community composition for phytoplankton/zooplankton groups	We added a simple sediment model, but it didn't make much difference for our 2km model of the BC coast - we wonder if it may be more impactful in higher resolution domains.	Holdsworth Christian
ECCC-CCMa	CanOE, CMOC	Global modelling (CanESM), Regional modelling/downscaling (CanTODS)	CanESM : Finalizing the requirements for the CMIP7 project and starting production experiments.	Jim Christian Krysten Rutherford Elise Olson
Dalhousie	bio_Fennel (ROMS)	Dalhousie's Marine Environmental Modelling Group		Katja Fennel
DFO (Pacific)	FVCOM-ICM	Fjords modelling (oxygen, submerged aquatic		Laura Bianucci

		vegetation)		
DFO (Pacific)	ROMS-BG CM			Angelica Pena
DFO (Pacific)	CanOE, CMOC	Hindcasts and projections of the Canadian Arctic; sea-ice bgc		Nadja Steiner

BGC Model Dev and Directions

1 slide updates

- CSIB2 (Antoine Haddon, Tessa Sou, Nadja Steiner)
 - Variable ice algal stoichiometry
 - Biomass export to ocean due to sea ice warming
 - Implementation to NEMO4.2-SI3
- BLING & BiGCIIIM (Inge Deschepper)
 - Upgrading to NEMO5 and SI3
 - Tidal and river runoff investigations
- NAODS-CanOE (Yujuan Sun, Diane Lavoie, Joel Chasse, Guoqi Han DFO)
 - NEMO3.6 CanOE to assess the impact of variability on ecosystems in NW Atlantic
 - Hindcast (MOI_BIO4) and climate simulations (MPI-ESM1-2-HR)
- 1D Coupled Physical-BGC Model (Jared Penney, Changheng Chen, Nancy Soontiens, Frederic Cyr, David Belanger, Aaron Adamack DFO)
 - Spring phytoplankton bloom on Newfoundland Shelf
 - Changes in ice season timing are shifting spring bloom time and relationship between the two
 - Aim to develop forecast model of spring bloom to help fisheries
- Extremes in NE Pacific (Amber Holdsworth DFO)
 - Unsupervised clustering for characterising ocean extremes (O₂, temp, acidification)
 - NEP36 CanOE, (NEMO3.6) 1996-Present
 - Recent work to add oxygen diagnostics to determine physical and bio controls on hypoxia
- CCCma CanTODS + BGC (Elise Olsen CCCMa, ECCC, DFO, Ualberta +)
 - CanTODS downscales from global CanESM, work being done to test/configure BGC options (CMOC, CanOE)
- SMELT (A. Tall, Susan Allen, UBC)
 - Improved O₂ by bathymetry updates
- FVCOM-ICM (Thomas Guyondet, DFO Gulf Region)
 - NPZD + O₂ coupled with TSS, benthic primary producers, bivalves
 - Scenarios: nitrate loading reduction, and bivalve mass mortality
 - Experimenting with NEMO coupling to a bivalve module
- Diane Lavoie

- Habitat characterisation for different species, collaboration with provincial govt,
- Gulf of St. Lawrence model to NEMO4.2: improvement of sediment, atm deposition of nitrogen, CMIP6 output to be used
- Krysten Rutherford (CCCMA at ECCC)
 - BGC in CanESM, 2 BGC models in the global
 - 2 active versions of CanESM (NEMO3.6) CanESM6 with NEMO4.2
 - Experiment from CMIP6:
 - Alkalinity to surface ocean, 60S-70N, using a high emission pathway
 - Comparing results with other Earth system models that contributed
- Jose Valenti (DFO SMELT)
 - SalishSeaCast updates: using Olsen 2020 version of SMELT, new scheme with Het. Bacteria

Intro for BGCM discussion

- Stated priorities for Canadian government and funding organisations (see slide for full details)
 - Climate change impacts on ecosystems
 - Earth system modelling of carbon cycle (particularly in northern regions)
 - Nature-based solutions incl CO₂ removal
 - Identifying vulnerable ecosystems and coastal regions
 - Advancing predictions and projections for extreme events (warming, acidification, deoxygenation)
 - Blue economy
 - Fast-emerging technologies (AI, ML)
 - Management decisions related to adapting fisheries, oceans and coastal infrastructure

Development needs to advance topics:

1. BGC model advances to accurately simulate response to extreme events and climate change
- Constraint of rate responses (remineralisation, primary production, grazing, mortality - to temperature and ocean acidification (OA))
 - Diane - Impact of OA on phytoplankton experiments
 - Amber - responses are not constrained through observations and experiments
 - Example of Salish Sea projections that Natasha was looking at: diatoms in Salish Sea seem to have an upper temp limit (~25 degrees), but limited confidence in that number, and when we project into future there are hesitations with interpretation
 - Need better rep of their functional forms, not only rates
 - Inge - have we been able to communicate with observationalists about this?

- Amber - not directly, but Susan's group has to some extent. Often observationalists require a level of complexity that is simply not available in models
- Inge - experience with observationalists similar; model can only do so much
- Diane - even if field observations show trends, we may not understand the underlying mechanisms
- Nadja - link to paper for sea-ice BGC modelling, what do we need as modellers from observationalists
- Antoine - experience working with observationalists is very beneficial, takes time to establish a common language,
 - Example: nitrates in sea ice can vary substantially even in one day, depends on when you take measurements
- Amber - capacity is more the issue for development advancement
- Nadja - need merging of observationalists and modellers, maybe creation of working groups with a modelling focus but include observationalists every few meetings
- Thomas - trying to push integration between modelling and observations
- Nadja - from Bepsii community: when we do data coalitions (?) can we add information on how this could be linked to modelling
- Jim - synthesis modelling project: process study cruises gathered a lot of information from observationalists and given to modellers without consultation on what modelling needs are
- Jim - NSERC grant selection committee - selection committees are very broad, a lot of the questions we need answered would be difficult to get through the selection process
- Elise - what is the role of adaptation on different time scales in determining effects, for ex: what timescales would evolutionary adaptation lead to changes in rate responses? A lot of research is going into this area right now, is there a way to leverage this in the modelling world?
- Jared - are people still doing experiments on phytoplankton growth rates? Often data is based on observations from long ago?
- Elise - mesocosm experiments exposing organisms to different treatments
- Inge - ULaval do experiments but are dependent on phytoplankton they can keep alive in the lab. Shift to light responses rather than temperature responses
- Diane - Martine Lizotte is doing experiments, collecting water in the field and then doing lab experiments
- Nadja - quite a bit of that work going on: temperature, OA, etc, these are what's being presented at conferences these days, on-ship experiments, OA in-lab experiments
- Jim - how can we leverage how the funding system works in Canada
- Paul - adding that this is no different on the physical oceanography side. One challenge is different funding pools for academics versus government. Have different levels of scientists collaboration - students working in government labs/offices

- Inge - BGC modelling groups are disparate - more communication between people modelling BGC would be helpful, linking people working on smaller regional models
- Nadja - biodiversity funding goals, but we lack this in our models; could be another avenue for collaboration between observationalists and modellers, even if we can't explicitly model biodiversity, there are ways we can work towards representing it
- Elise - this is a big struggle with the BGC modelling we do
- Thomas - find ways to model biodiversity

2. BGCM with tides

- Incorporating tidal effects in global/basin-scale models?

3. Diversity of BGC models:

- Paul - diversity of BGC models: we have BLING at UofA, SalishSeaCast modules, CanOE, PISCES in the NEMO space, in the physical space, we have a number of models being run in Canada, do we have too much diversity in configurations with a limited number of models? Is this improving our science or making it harder to do BGC modelling in Canada?
- Elise - In regional domains, there's benefit to having a model fit for that region; in the context of downscaling, when you just move a global model to a coastal region, the resolution isn't sufficient for the accuracy needed; sometimes the models need modifications for rates or phytoplankton groups
- Nadja - the less constrained we have, the more diverse our solutions become; for BGC, many models even used in ESMs are developed somewhere locally and may not be relevant for other regions
- Jim - ex: most modelling centres are going to tune their models for their location, need objectivity for these tuning parameters.
- Inge - BGC tuning is behind the physics-based tuning; we don't have the same advantage of time (50 years of experience) that the physics have

4. Constraining BGC parameters with respect to river input

- Inge - UManitoba, work beginning on watershed inputs, seasonal-monthly scale, apply to models
- Paul - on the physical side, working with T. Stadnyk at UCalgary who does hydrological modelling, AHype model, pan-Arctic rivers, distributed model of rivers, lakes, dams, precipitation, river temp, talking with group in Sweden about adding in a BGC model to this
 - Joint proposal for a pan-Arctic modelling systems; post-doc position coming if funding is successful
- Krysten - question: thoughts on AI/ML in ocean BGC space?
 - We don't necessarily have sufficient training data for this, but could it be helpful?

- Thomas - make the link with the river aspect; one of the fields where you could use ML methods is input generation, ex: hydrological modelling to provide nutrient loading into the system
- Heather - Mercator Ocean International has been exploring ML in BGC modelling
 - They've been developing BGC hindcasts assimilating ocean colour to support that.
 - Coralie Perruche contact

Papers and resources:

- Archibald 2022,
- Nadja's sea ice BGC model paper:
<https://online.ucpress.edu/elementa/article/doi/10.12952/journal.elementa.000084/112911/What-sea-ice-biogeochemical-modellers-need-from>
- Oxygen improvements on SalishSeaCast:
<https://essopenarchive.org/doi/full/10.22541/essoar.177100396.68261433/v2>
- GLOBAL NEWS 2: <https://gmd.copernicus.org/articles/19/867/2026/#section3>
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5. Cross-institute coordination of NEMO development

(Discussion leads: Natasha Ridenour and Heather Andres)

Wednesday 13 May, 12:00 pm MDT

Natasha- Challenges and Motivation (see Workshop slides for details)

- Each group has put in effort and has deliverables
- Different resolutions and/or different model components
- Funding is hard to obtain
- Not enough people or time/resources
- Generally, all have the same NEMO problems
- Generally have the same goal of better models and results
- Examples of coordination:
 - CanTODS built off of work at DFO and ECCC
 - Code sharing between ECCC and DFO
 - UofM analysis package development
 - UBC and DFO (IOS) regular monthly meetings
 - CONCEPTS
 - DFO and ECCC memoranda of understanding
- We should advertise the website more
 - Pages for institutions
 - Great place to get to know who people are, especially with government restrictions
 - Can add "works in progress" section per institute. Help advertise and bridge gap of connecting people

Heather-

- How do we subdivide the sharing on different levels for different needs

Discussion questions-

- What are the limiting factors...?

- Paul: Forcing. Is a 30km res atm forcing for a high-resolution model adequate? River runoff, melt from ice sheets. Evaluation: if you switch versions, how do you adjust it? People power. How to train and keep people?
- Stephanie: Hard to hire people for extended work. See benefit in having people around for a while.
- Guoqi: Difficult to find motivated and skilful scientists who will sit down and improve the model. Developing new capacity in the model. Recruitment and funding are not attracting these skilful modellers. If parameterisation adjustments/adaptations can be well documented and made available (stuff that won't be a paper), it needs to be communicated and can contribute to the community
- Natasha: Recruiting and keeping people is difficult, but is it also with people that are applying and a gap in their skills that you are looking for? Is there a gap in their skill sets, or is it that not many people are applying to the jobs?
- Guoqi: Some skills gap in the skill set and set high standards. May lack leadership but is good at modelling. It is the combination of all qualities and not just one focus.
- Stephanie: Even people with the skills, there is no option to offer only a casual contract too; it is hard to keep them around and waiting for a permanent position. Hard to cultivate the skills if only short term. Students need to hustle and pay rent. Can't put motivation into work if needing to survive. Don't see it changing in government soon.
- Fred: If we ignore the hiring issue in the last years, it was hard to find true modellers. We are having a high loss rate of true modellers. Short contract times are hard to navigate. Brain drain in Canada?
- Paul: Academia cannot absorb all the modellers and retain them. In Academia, we want to train and then get them to leave for their next position. We don't pay our grad students enough. Not great or stable. Postdoc is transient and short-term. The whole system has issues with attracting and retaining the best people. It is based on the opportunities.
- Heather: Communication of what is happening in government in PhD was not good.
- Natasha: Improve communication that the government does do science and that they hire. CMOS is a current platform to use.
- Heather: Partnering between government and academia when students are students? Can it open communication and doors?
- Natasha: Some links with Paul and CCCma.
- Paul: There are links with uVic and IOS and Dal and DFO(? check). Getting it formally set up is the challenge. For student defences, but the government doesn't supervise so the university says no.
- Stephanie: A note on student funding: the dollar amount of NSERC graduate student awards (CGRS / PGRS) has only just been increased to a few thousand more than what I was awarded ~10-15 years ago -- it absolutely has not kept up with the cost of living.

- Guoqi: Get government scientists as adjunct professors to supervise and join projects across different agencies.
- Heather: There are barriers now as government scientists can not ask to be adjunct professors, they have to be invited.
- Nicolas: CP4C is a collaborative platform for the university to use the CanESM model system, and it has been used in many places: McGill, University of Toronto, etc. [CP4C - Collaborative Platform for CanESM](#)
- Inge: had co-supervision during PhD and it was very beneficial.
- Natasha: Young government researchers need to explore being adjuncts.
- Possible areas of sharing/coordination
 - Fred: Model code is useful for learning from one another and for sharing as much code as possible.
 - Natasha: Just having the code visible? Or being aware of different issues and how they were solved?
 - Fred: Information on the code modifications. We share code from the government internally (DFO), and hard to make it public.
 - Nicolas: mirrored repository for CCCma
 - Fred: Potentially more time when leaving the CICE consortium.
 - Nicolas: Git is a very useful tool. Many capabilities. Can tag people and discuss questions.
 - Hatem: Is there space on the lines of ML?
 - Natasha: model parameterisation/ adaptation?
 - Debora: Regarding the model domain, high resolution in coastal areas would be greatly appreciated. So we could better predict the effects of climate change on this economically important area.
 - Hatem: ensemble and stochastic simulations approach
 - Stephanne: Fewer people are doing ensembles as it is computationally expensive.
 - Heather: Rivers?
 - Natasha: CCCma work may have 3-ocean river data but won't be appropriate for higher-resolution domains.
 - Stephanne: How do we make several TB's of data available, or what is useful for people?
 - Hatem: Yes, it is useful. High-resolution data is important. But making it available is the question.
 - Stephanne: Is it okay to have just the geometry information or more?
 - Hatem: The data is important.
 - Heather: Which is most doable vs the amount of work that would go into achieve it? Bang for buck!
 - Stephanne: Diagnostic analysis tools are low-hanging fruit. Obs datasets are many times not ours to share and have formatting issues. Namelists are text files that can easily be shared. Domain and config, data, where to store it?
 - Paul: Obs datasets are smaller, but the format for model evaluations is more difficult. We are not the experts for processing observations to get it in the format that can be useful for us. Need to talk with obs communities

more. Depending on what you get from different databases, can I pull out information from the time I need? More challenging than we think.

- Stephanie: The processing of the observations is time-consuming.
- Paul: Talking with relevant scientists, we can get a good dataset, but it is not a common occurrence.
- Nicolas: Low-hanging fruit for git coordination (organisation): share our existing repository with the group. A neutral place that can host it. Doesn't have to be just about code. Can add a page for obs or analysis tools...
- Stephanie: Shorter name for the website?
- Elise: What if the format of the monthly seminars are adjusted with one presentation and then the other half an open discussion session?
- Tahya: Try to have 2 talks related to a theme. So keeping it related to a theme would help so that it varies per month.
- Paul: I don't think there are any rules for the monthly seminars. The hardest part is finding volunteers to lead discussions. It can be accommodated.
- **Steps forward**
 - Share a git repo
 - Nicolas- Our group is using GitLab 😊 [CCCma · GitLab](#)
 - This is the GitHub project for the documentation about the Canadian Community of practice:
[UofA-NEMO/Canadian-NEMO-Ocean-Modelling-Forum-Community-of-Practice](#)
 - GitHub offers a free organisation. Would there be an issue with that, and why not have an academic institute host it?
 - Forum email list- requires work to maintain
 - Suggestion of a quarterly meeting, but if there are discussions in the monthly seminars, then do we need a quarterly meeting?
 - Amber: Examples of things that have been tested can be added to the GitHub. There are internal reports that can be shared. Promise to put up on the website or on GitHub.
 - Natasha: No need for extreme details, but a brief comment about what was done and why with the before and after examples.
 - Amber: People can then add comments to each thing to add discussion between groups
 - Michael: Nemo forum material. <https://nemo-ocean.discourse.group/>
 - Heather: Is there a reason to improve and focus on something that is just Canadian or link to the NEMO forum?
 - Nicolas: We should ask more questions on the NEMO forum. Direct technical questions there help to increase our visibility in the international community too.
 - Stephanie: Visibility is good in international forums; it works for NEMO, but if we develop things that are on top of NEMO, but not NEMO specific it will be excluding sections of the community.
 - Natasha: That would lock out a lot of people that are doing BGC modelling that is not PISCES

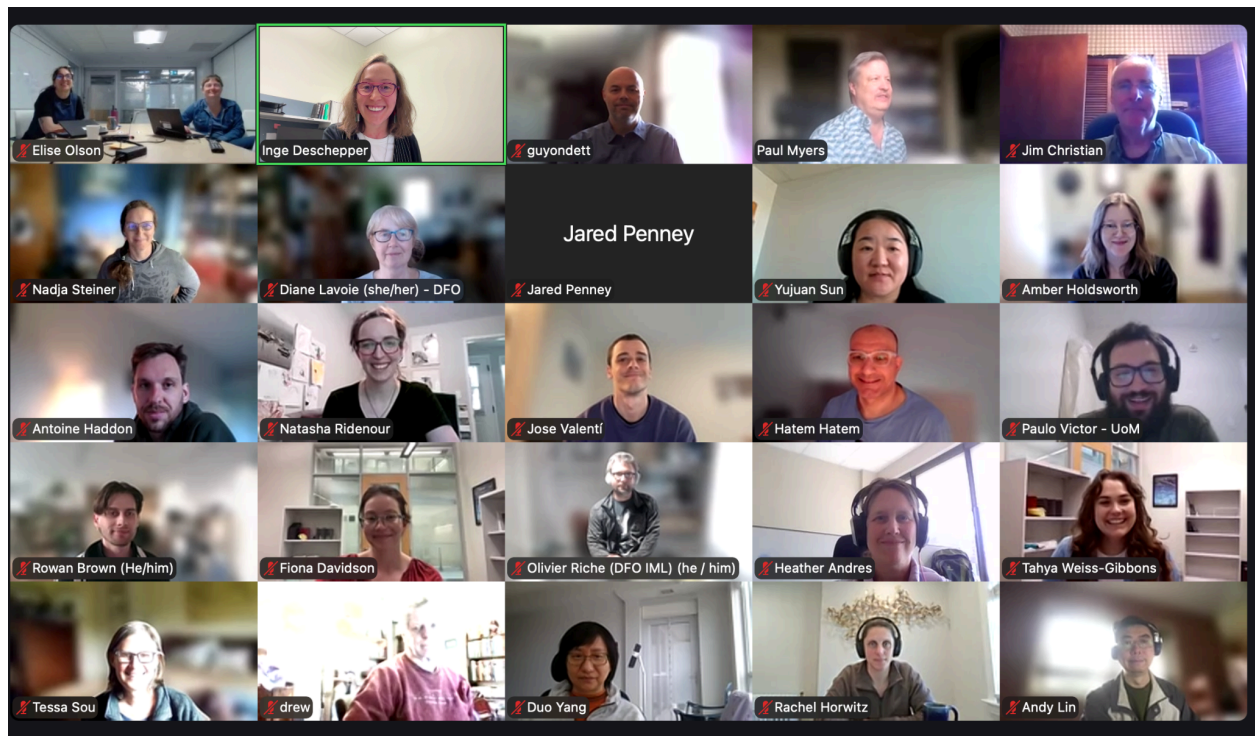
- Amber: Some things that are specific to modelling in Canada, the git could be more useful than the NEMO forum
- Natasha: Within the parameter space, is there a way to share the load of parameter testing? There seems to be a lack of people and time. And communicate which tests have been done so that they are not repeated.
- Paul: Advantage for ECRs to have a Canadian-based discussion. Jobs, opportunities and communication.
- Natasha: Less hesitant to post on a Canadian forum than just the NEMO forum.
- Heather: A good place to start is this Git. There is enough work to make it beneficial for the Canadian NEMO forum. 1st Action item (create a GitHub setup). Who would set it up? Once it is set up, can you add something to test and see how it will work?
- Jared: not volunteering. Is there a preference between GitLab or GitHub?
- Nicolas: GitHub organisations can be free, but GitLab is limited to the number of people who can collaborate.
- Natasha: Can contribute to submissions on GitHub.
- Paul: GitHub setup will be through UofA
- Tahya: Can help
- Natasha: Once setup, send to mailing list
- Nicolas: Can help Natasha with Git. There is an easy way to migrate from different systems
- Thomas: Parameter testing questions, would simple test cases be useful? Simple geometry, with code and inputs shared? Can be a couple of cases.
- Amber: Could this be a good place to add information to the website to state what is available? Start a list for the GitHub or website.
- Paul: The website can have information on configurations, experiments, and namelists. It is possible to use it to supply this information.
- Geoff: CCCma, we have a scheme to adaptively change the use of the GM parameter. This information is available on the GitLab of the CCCma. Can add this information onto the forums GitHub.
- Heather: Please add information on websites and update it. Volunteers that will contribute to the GitHub
- Amber: Thought will have to go into how it will look and be structured. Potentially, Nicolas could help guide us on how it will look and be structured.
- Nicolas: Start with a prototype with Paul, go through rounds of comments and changes, and then release it. Please have the community commit to using it.
- Amber: If we have examples, we could have them explained at the seminars to promote usage and git usage.

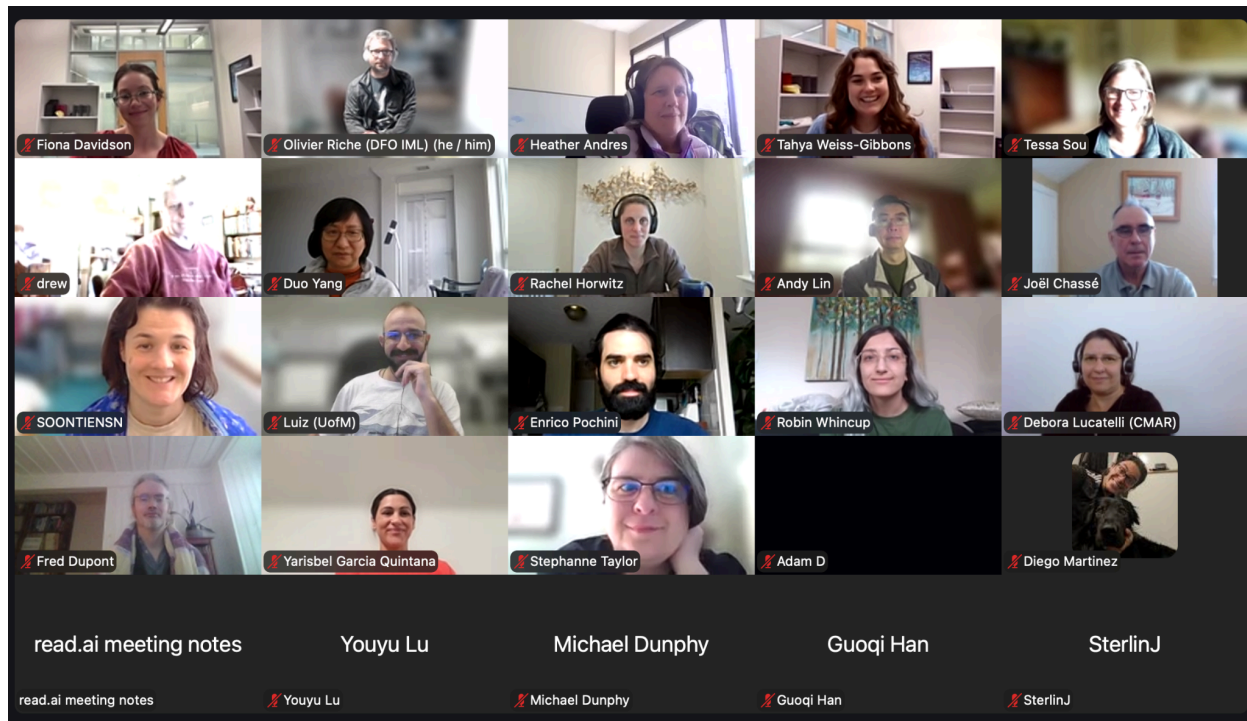
Papers and resources:

6. Wrap-up (Paul Myers)

Paul- There are many common topics and areas that need to be discussed.

- Need to have more links with CIOOS and potentially link with a seminar
- Lead to a white paper on Ocean modelling in Canada. Needs to move forward
- Last year it was a hybrid day. Online this year. Aim for a hybrid event next year to encourage ECR discussions
- Would it be worthwhile to have a shorter discussion in 6 months, or will the seminars with discussions suffice?
- Please volunteer for seminars
- We have ECR travel funding available. Please apply!





Day 3 participants from left to right:

Kyrsten Rutherford, Elise Olson, Inge Deschepper, Thomas Guyondett, Paul Myers, Jim Christian, Nadja Steiner, Diane Lavoie, Jared Penney, Yajuan Sun, Amber Holdsworth, Antoine Haddon, Natasha Ridenour, Jose Valenti, Hatem Hatem, Paulo Victor, Rowan Brown, Fiona Davidson, Olivier Richie, Heather Andres, Tahya Weiss-Gibbons, Tessa Sou, Drew (?), Duo Yang, Rachel Horwitz, Andy Lin, Joël Chassé, Soontiensn, Luiz Henrique de Silva, Enrico Pochini, Robin Whincup, Debora Lucatelli, Fred Dupont, Yarisbel Garcia Quintana, Stephanie Taylor, Adam D, Diego Martinez, Youyu Lu, Michael Dunphy, Guoqi Han, J. Stelin.